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1. (a) What is the smallest positive integer n such that there are exactly four non-isomorphic Abelian groups of order n ? Name the four groups.

By the Fundamental Theorem of Finite Abelian Groups, we want to find a prime p such that the number of partitions of p is 4. Consider the product $p_1^2 p_2^2$ where p_1 and p_2 are distinct primes. Since both amount to 2 partitions respectively, their product amounts to 4 partitions. To get the smallest integer n , we want to use the smallest primes that create 4 partitions. So,

$$n = 2^2 \cdot 3^2 = \boxed{36}$$

The groups should be of the forms $\mathbb{Z}_{p_1^2} \oplus \mathbb{Z}_{p_2^2}$, $\mathbb{Z}_{p_1^2} \oplus (\mathbb{Z}_{p_2} \oplus \mathbb{Z}_{p_2})$, $(\mathbb{Z}_{p_1} \oplus \mathbb{Z}_{p_1}) \oplus \mathbb{Z}_{p_2^2}$, and $(\mathbb{Z}_{p_1} \oplus \mathbb{Z}_{p_1}) \oplus (\mathbb{Z}_{p_2} \oplus \mathbb{Z}_{p_2})$. Thus, the groups are

1. $\mathbb{Z}_4 \oplus \mathbb{Z}_9$
2. $\mathbb{Z}_4 \oplus (\mathbb{Z}_3 \oplus \mathbb{Z}_3)$
3. $(\mathbb{Z}_2 \oplus \mathbb{Z}_2) \oplus \mathbb{Z}_9$
4. $(\mathbb{Z}_2 \oplus \mathbb{Z}_2) \oplus (\mathbb{Z}_3 \oplus \mathbb{Z}_3)$

- (b) Up to isomorphism, how many additive Abelian groups of order 16 have the property that $x + x + x + x = 0$ for all x in the group?

Since 2 is prime and $2^4 = 16$, we want to find the groups corresponding to the partitions of p^4 . Since partitions of p^4 are p^4, p^3p, p^2p^2, p^2pp , and $pppp$, our choices of groups are

1. $\mathbb{Z}_{16}$ (max order of 16)
2. $\mathbb{Z}_8 \oplus \mathbb{Z}_2$ (max order of 8)
3. $\mathbb{Z}_4 \oplus \mathbb{Z}_4$ (max order of 4)
4. $\mathbb{Z}_4 \oplus (\mathbb{Z}_2 \oplus \mathbb{Z}_2)$ (max order of 4)
5. $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$ (max order of 2)

Groups whose elements have a max order of 4 satisfy the property. Also, groups with max order of 2 also satisfy it since $(x + x) + (x + x) = 0 + 0 = 0$. So, 3 groups satisfy the property.

- (c) Let $p_1, p_2, \dots, p_n$ be distinct primes. Up to isomorphism, how many Abelian groups are there of order $p_1^4 p_2^4 \cdots p_n^4$?

For each p_i^4 , it has 5 partitions. So, the total number of groups of this order is 5^n

2. Let $G = \{1, 7, 43, 49, 51, 57, 93, 99, 101, 107, 143, 149, 151, 157, 193, 199\}$ under multiplication modulo 200. Determine the isomorphism class of G . *Justify your answer.*

Since $|G| = 16$, we want to find E.D.Ps of order 16. Since $2^4 = 16$, the only options are

$\mathbb{Z}_{16}$	possible orders: 1, 2, 4, 8, 16
$\mathbb{Z}_8 \oplus \mathbb{Z}_2$	possible orders: 1, 2, 4, 8
$\mathbb{Z}_4 \oplus \mathbb{Z}_4$	possible orders: 1, 2, 4
$\mathbb{Z}_4 \oplus (\mathbb{Z}_2 \oplus \mathbb{Z}_2)$	possible orders: 1, 2, 4
$\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$	possible orders: 1, 2

The orders of the elements in G under multiplication modulo 200 are:

$$\begin{aligned}
 |1| &= 1, & |7| &= 4, & |43| &= 4, & |49| &= 2, \\
 |51| &= 2, & |57| &= 4, & |93| &= 4, & |99| &= 2, \\
 |101| &= 2, & |107| &= 4, & |143| &= 4, & |149| &= 2, \\
 |151| &= 2, & |157| &= 4, & |193| &= 4, & |199| &= 2
 \end{aligned}$$

Since no elements have order 8 or 16, we can eliminate $\mathbb{Z}_{16}$ and $\mathbb{Z}_8 \oplus \mathbb{Z}_2$. Also, since there's at least one element of order 3, we can eliminate $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$. So, the only options left are $\boxed{\mathbb{Z}_4 \oplus \mathbb{Z}_4 \text{ and } \mathbb{Z}_4 \oplus (\mathbb{Z}_2 \oplus \mathbb{Z}_2)}$

3. Prove Theorem 12.1, which says the following:

Let a , b , and c belong to a ring R . Then

1. $a0 = 0a = 0$

Proof. $a0 = a(0 + 0) = a0 + a0$. Adding $(-a0)$ to both sides,

$$\begin{aligned}
 a0 + (-a0) &= (a0 + a0) + (-a0) \\
 \Rightarrow 0 &= a0
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 0a &= (0 + 0)a = 0a + 0a \\
 \Rightarrow 0a + (-0a) &= (0a + 0a) + (-0a) \\
 \Rightarrow 0 &= 0a
 \end{aligned}$$

□

2. $a(-b) = (-a)b = -(ab)$

Proof. If $a(-b) = -(ab)$, then that implies $a(-b)$ is the additive inverse of ab . So,

$$a(-b) + ab = a(-b + b) = a0 = 0.$$

So,

$$\begin{aligned} a0 + (-(ab)) &= 0 + (-(ab)) \\ \Rightarrow a(0 + (-b)) &= -(ab) \\ \Rightarrow a(-b) &= -(ab) \end{aligned}$$

Similarly,

$$\begin{aligned} (-a)b + ab &= (-a + a)b = 0b = 0. \\ \Rightarrow 0b + (-(ab)) &= 0 + (-(ab)) \\ \Rightarrow (0 + (-a))b &= -(ab) \\ \Rightarrow (-a)b &= -(ab) \end{aligned}$$

□

3. $(-a)(-b) = ab$

Proof. By Property 2 of Theorem 12.1,

$$\begin{aligned} (-a)(-b) &= -((-a)b) \\ &= -(-(ab)) \\ &= ab \end{aligned}$$

□

4. $a(b - c) = ab - ac$ and $(b - c)a = ba - ca$

Proof. By the definition of rings and Property 2 of Theorem 12.1,

$$\begin{aligned} a(b - c) &= a(b + (-c)) \\ &= ab + a(-c) \\ &= ab + (-(ac)) \\ &= ab - ac \end{aligned}$$

Similarly,

$$\begin{aligned} (b - c)a &= (b + (-c))a \\ &= ba + (-ca) \\ &= ba - ca \end{aligned}$$

□

Furthermore, if R has a unity element 1, then

5. $(-1)a = -a$

Proof. By Property 2 of Theorem 12.1,

$$\begin{aligned}(-1)a &= -(1a) \\ &= -(a) \\ &= -a\end{aligned}$$

□

6. $(-1)(-1) = 1$

Proof. By Property 3 of Theorem 12.1,

$$\begin{aligned}(-1)(-1) &= (1)(1) \\ &= 1\end{aligned}$$

□

4. (a) Find an integer n so that the ring $\mathbb{Z}_n$ does **not** satisfy any of the following properties for all $a, b, c \in \mathbb{Z}_n$:

- i. If $a^2 = a$, then $a = 0$ or $a = 1$
- ii. If $ab = 0$, then $a = 0$ or $b = 0$
- iii. If $ab = ac$ and $a \neq 0$, then $b = c$.

Let $n = 6$ and consider the ring $\mathbb{Z}_6$:

- i. $3^2 = 9 \bmod 6 = 3$, but $3 \neq 0$ and $3 \neq 1$.
 - ii. $(3)(2) = 6 \bmod 6 = 0$, but $3 \neq 0$ and $2 \neq 0$.
 - iii. $(3)(4) = 12 \bmod 6 = 0$ and $(3)(2) = 6 \bmod 6 = 0$, but $4 \neq 2$.
- (b) Show that the three properties in part (a) are satisfied in $\mathbb{Z}_p$, where p is prime.

Since $\mathbb{Z}_p$ has a unity of 1, it has multiplicative inverses. So,

i.

$$\begin{aligned}a^2(a^{-1}) &= a(a^{-1}) \\ \Rightarrow a &= 0.\end{aligned}$$

If $a \neq 0$, then

$$\begin{aligned}a^2 &= a \\ \Rightarrow a^2 - a &= 0 \\ \Rightarrow a(a - 1) &= 0 \\ \Rightarrow a^{-1}(a(a - 1)) &= a^{-1}0 \\ \Rightarrow a - 1 &= 0 \\ \Rightarrow a &= 1\end{aligned}$$

ii.

$$ab(b^{-1}) = 0(b^{-1})$$

$$\Rightarrow a = 0$$

or

$$a^{-1}(ab) = a^{-1}0$$

$$\Rightarrow b = 0$$

iii.

$$(a^{-1})(ab) = (a^{-1})(ac)$$

$$\Rightarrow b = c$$

5. (a) Give an example of ring elements a and b with the property that $ab = 0$, but $ba \neq 0$.

(b) Let n be an integer greater than 1. In a ring in which $x^n = x$ for all x , show that $ab = 0$ implies that $ba = 0$.

(a) If $ab = ba = 0$, that means a and b commute. So, we want to find ring elements that don't commute. Consider the ring $M_2(\mathbb{Z})$ and let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ be elements of the ring. Then,

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

So, $AB \neq BA$.

(b) Let $n = 2$. Then,

$$(ba)^2 = ba(ba)$$

$$= b(ab)a$$

$$= b(0)a$$

$$= 0$$

By the definition of the ring, $(ba)^2 = ba$. So, $ba = 0$.

6. Let a be a fixed element of a ring R , and let $S = \{x \in R \mid ax = 0\}$. Show that S is a subring of R .

Proof. Let R be a ring, a be a fixed element of R , and $S = \{x \in R \mid ax = 0\}$. By the Subring Test, we need to show that S is closed under subtraction and multiplication. Take

$x, y \in S$ and consider the product $a(xy)$. By Theorem 12.1,

$$\begin{aligned} a(xy) &= (ax)y \\ &= 0y \\ &= 0 \end{aligned}$$

So, $xy \in S$. Similarly,

$$\begin{aligned} a(x - y) &= ax - ay \\ &= 0 - 0 \\ &= 0. \end{aligned}$$

So, $x - y \in S$. By the Subring Test, S is a subring of R .

□